



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
FULL SYLLABUS
13-04-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE : [SCORE (ALL STAR BATCH)]

PAPER-1

Time : 3 Hours

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

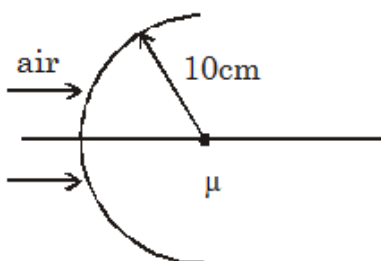
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

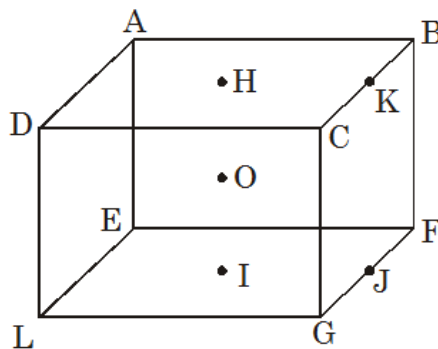
1. A spherical surface separates air & medium for which $\mu = 1.615$ for violet and $\mu = 1.600$ for red color. A paraxial beam parallel to optic axis is incident on the surface as shown. The distance between point of convergence for violet and red color is Δf .



- (A) $\Delta f = 0.41$ cm
 (B) Point of convergence for red is closer to optical centre than that for violet.
 (C) Point of convergence for violet is closer to optical centre than that for red.
 (D) $\Delta f = 0.84$ cm.

Space for Rough Work

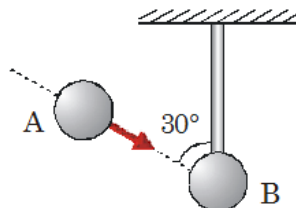
2. A cubic frame is made of 12 rods each of mass m and length ℓ .



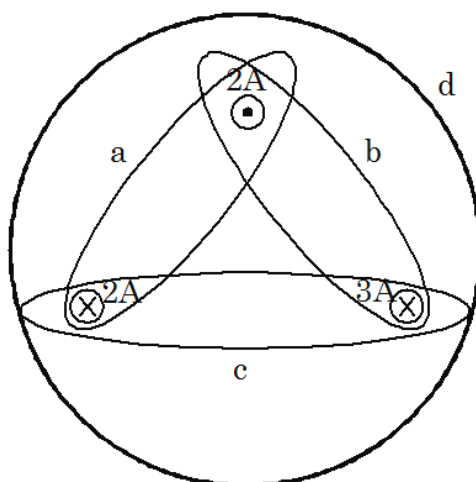
- (A) The moment of inertia of the cube about its face diagonal (BD) is $\frac{23}{3}m\ell^2$.
- (B) The moment of inertia of the cube about an axis passing through the center of cube and perpendicular to one of the face (HI) is $\frac{14}{3}m\ell^2$.
- (C) The moment of inertia of the cube about one of the side of cube (AB) is $\frac{32}{3}m\ell^2$.
- (D) The moment of inertia of the cube about an axis passing through the center of the face and the midpoint of the one of the side (JK) is $\frac{23}{6}m\ell^2$.

Space for Rough Work

3. A smooth sphere of mass 1 kg is tied to a fixed point by an inextensible string and suspended from the roof. Another identical sphere impinges directly on it with speed 5 m/sec in a direction making an acute angle 30° with the string. Co-efficient of restitution is $\frac{1}{2}$.



- (A) Speed of sphere B just after collision is 3 m/s.
 (B) Speed of sphere A just after collision is 2 m/s.
 (C) Impulse due to normal force between two spheres during collision is 6 N-S
 (D) Impulse due to tension in string during collision is 3 N-S.
4. Rank the magnitude of $\oint \vec{B} \cdot d\vec{\ell}$ for the closed paths shown in figure from the smallest to largest.



- (A) a, b, c, d (B) a, b, d, c (C) a, d, c, b (D) a, c, b, d

Space for Rough Work

5. A point charge q is placed at origin. The electrostatic energy stored in the space around the point charge in various volumes as mentioned in listed in options below. Choose the correct option(s) below.
- (A) A small cube with edge size a ($a \ll r$) placed at $(r, 0, 0)$ has $\frac{q^2 a^3}{32\pi^2 \epsilon_0 r^4}$ amount of energy.
- (B) A small cube with edge size a ($a \ll r$) placed at $(r, 0, 0)$ has $\frac{q^2 a^3}{16\pi^2 \epsilon_0 r^4}$ amount of energy
- (C) Energy stored in volume given by $a^2 \leq x^2 + y^2 + z^2 \leq 4a^2$ is $\frac{q^2}{16\pi \epsilon_0 a}$
- (D) Energy stored in volume given by $a^2 \leq x^2 + y^2 + z^2 \leq 4a^2$ is $\frac{q^2}{32\pi \epsilon_0 a}$
6. A string of mass 0.2 kg and length 2m is tied at two ends to fixed supports under a tension of 10 N. A point P on the string is found to travel from one extreme to other in 0.1s. Taking one end as $x = 0$ and the other end $x = 2$ m and $t = 0$ as the time when P is at rest. (Position of P is x) The correct statements will be :-
- (A) For $0 < t < 0.1$ s, energy flows across P in positive x -direction for $0 < x < 1$ m
- (B) For $0 < t < 0.05$ s, energy flows across P in negative x -direction for $0 < x < 1$ m
- (C) At $t = 0.05$ s, rate of energy flow through P is zero for $x = 0.5$ m
- (D) At $t = 0.1$ s, rate of energy flow through P for all values of x is zero.

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

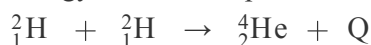
Paragraph for Questions 7 and 8

Nuclear binding energy varies from nucleus to nucleus, it is necessary to compare the binding energy per nucleon basis. If we find binding energy per nucleon for a given element it can be given as binding energy divided by the nucleon number A as

$$(BE)_N = \frac{\Delta E}{A} = \frac{\Delta mc^2}{A}$$

We can define binding energy per nucleon theoretically as the amount of energy needed to remove a nucleon from nucleus of an element.

7. The binding energy of deuteron (${}^2_1\text{H}$) is 1.15 MeV per nucleon and an alpha particle (${}^4_2\text{He}$) has a binding energy of 7.1 MeV per nucleon. Then in the reaction



the energy Q released is :

- (A) 1MeV (B) 11.9 MeV (C) 23.8 MeV (D) 931 MeV

8. What is the energy released in the fission reaction



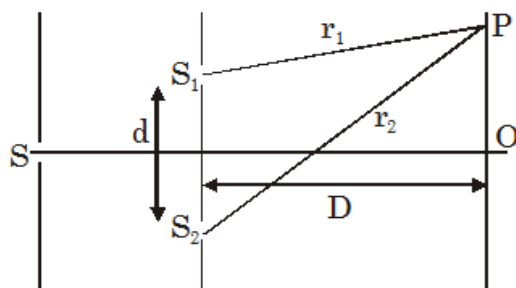
given that the binding energy per nucleon of X and Y is 8.5 MeV and that of ${}^{236}_{92}\text{U}$ is 7.6 MeV?

- (A) 176.2 MeV (B) 183.7 MeV (C) 195.4 MeV (D) 208.6 MeV

Space for Rough Work

Paragraph for Questions 9 and 10

An electromagnetic wave can be represented by $E = A \sin (kx - \omega t + \phi)$, where E is electric field associated with wave. According to this equation, for any value of x , E remains sinusoidal for $-\infty < t < \infty$. Obviously this corresponds to an idealised situation because radiation from ordinary source consists of finite size wavetrains. In general, electric field remains sinusoidal only for times of order τ_c , which is called coherence time. In simpler language it means that for times of order τ_c , a wave will have a definite phase. The finite value of coherence time could be due to many factors, for example if radiating atom undergoes collision with another atom then wave train undergoes an abrupt phase change or due to the fact that an atom responsible for emitting radiation has a finite life time in the energy level from which it drops to lower energy level, while radiating. Concept of coherence time can be easily understood using young's double slit experiment. Let interference pattern is observed around point P at time t , due to superposition of waves emanating from S_1 and S_2 at times $t = \frac{r_1}{c}$ and $\frac{r_2}{c}$ respectively, where r_1 and r_2 are the distance S_1P & S_2P . Obviously if $\frac{r_2 - r_1}{c} \ll \tau_c$, {where $c = 3 \times 10^8$ m/s} then, wavetrain arriving at point P from S_1 & S_2 will have a definite phase relationship and an interference pattern of good contrast will be obtained.



9. Which of the following statements is incorrect in the diagram given :-
- (A) Central fringe (near O) will always have a good contrast.
 - (B) As we move on the screen towards higher orders, contrast will become poorer.
 - (C) If we move S, parallel to S_1S_2 there will be no change in contrast of fringes on screen around O.
 - (D) If we move on screen away from O, fringes will disappear beyond a certain limit.
10. If coherence time is of order 10^{-10} second and screen is placed at a very large distance from slits in the given figure, then :-
- (A) For $d = 0.3$ cm, we will not get any fringes on screen (except central maxima)
 - (B) For $d = 0.03$ cm we will get fringes on a large part of screen.
 - (C) For $d = 3$ cm we will get fringes on entire screen.
 - (D) We will get fringes on entire screen, irrespective of value of d .

Space for Rough Work

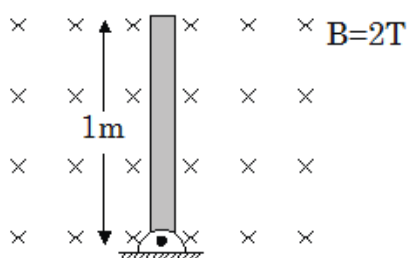
SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

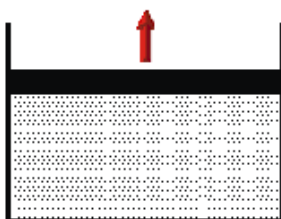
1. A rod of length 1m rotates about one of its end points with an angular velocity 2 rad/sec in a plane perpendicular to the magnetic field $B = 2\text{T}$ as shown in the figure. Then find magnitude of electric field (in SI unit) at the mid point of the rod.



2. A steel rod at 25°C is bolted at both ends and then cooled. By how many $^\circ\text{C}$ should the rod be cooled so that it will rupture? Assume that till rupture, Hooke's law is obeyed. ($\alpha_{\text{steel}} = 10 \times 10^{-6} \text{ }^\circ\text{C}^{-1}$, $Y = 2 \times 10^{11} \text{ N/m}^2$ and σ_b (breaking stress of steel rod) $= 4 \times 10^8 \text{ N/m}^2$)

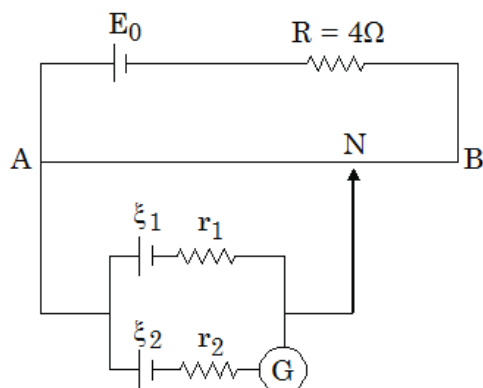
Space for Rough Work

3. The velocity distribution in a viscous flow over a plate is given by $u = 4y - y^2$ for $y \leq 2\text{m}$ where u = velocity in m/s at a point distant y from the plate. If the coefficient of dynamic viscosity is 1.5 Pa.s . Determine the shear stress at $y = 1\text{m}$ (SI units)
4. In the figure shown water is filled in the cylindrical tank of cross-sectional area 100 cm^2 up to a height of 10 cm . The mass of the thin, tight fitting and frictionless piston is 100 kg . The small hole in the piston has area $2\sqrt{2} \text{ mm}^2$. Due to weight of the piston water spurts upwards in a jet from the hole. Find the time (in sec.) in which the tank becomes empty. (Assume that spurted water does not fall back on piston)

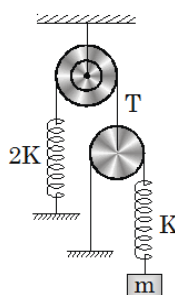


Space for Rough Work

5. A battery of emf $E_0 = 12.5 \text{ V}$ is connected across a 4 m long uniform wire having resistance $4\Omega/\text{m}$. Two cell of small EMFs $\xi_1 = 2\text{ V}$ and $\xi_2 = 4\text{ V}$ having internal resistance 2Ω and 6Ω respectively are connected as shown in the figure. If galvanometer shows no deflection at the point N, the distance of point N (in m) from the point A is equal to :-

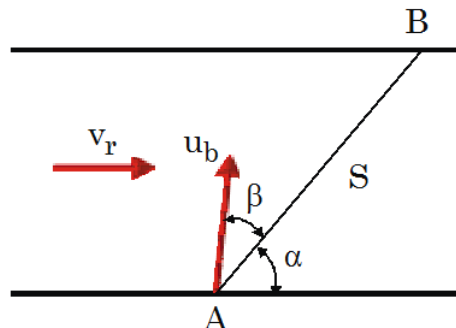


6. If the block shown in figure is displaced slightly and released, find the time period of oscillations of the block is found to be $2\pi\sqrt{\frac{Pm}{K}}$. Find P.

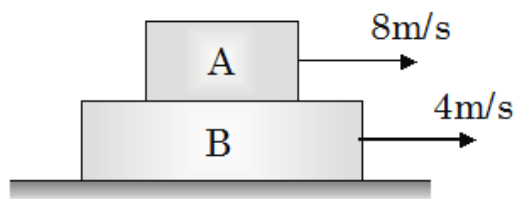


Space for Rough Work

7. A boat travel, between two points A and B on the opposite banks of a river (figure), always following the line AB. The distance S between points A and B is 240 m. The velocity of the river flow $v_r = 2\sqrt{2}$ m/s is constant over the entire width of the river. The line AB makes an $\alpha = 45^\circ$ with the direction of the current. The boat moves to cover the distance AB and back in a time $t = 2\sqrt{3}$ min. The angle β remains the same during the passage from A to B and from B to A. With what velocity u_b should the boat travel ?



8. At an instant $t = 0$, Block A of mass 1 kg is moving with speed 8 m/s towards right on rough surface of block B of mass 3 kg. Block B, which is placed on smooth horizontal surface is moving with speed 4 m/s towards right at same instant ($t = 0$). The net work done against the frictional force in long time is (block A will not fall from block B)



Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

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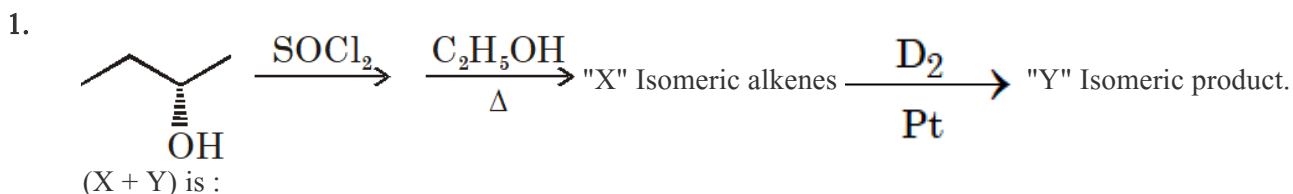
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

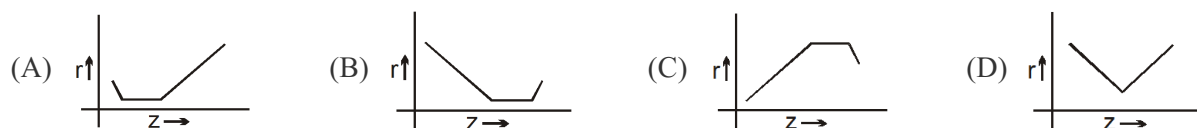
Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.



- (A) 3 (B) 2 (C) 4 (D) 8

2. Which is best graphical representation of atomic radius for 3d series elements.

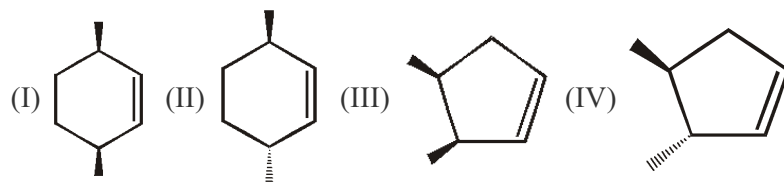


Space for Rough Work

3. Which is/are correct statements ?

- (A) AgBr shows Frenkel defect as well as schottky defect.
- (B) Density may be increase due to impurity defect.
- (C) End centred cubic unit cell does not exist in nature
- (D) Orthorhombic crystal system has three bravias lattices.

4.



Which of the reactant will formed optically active product after reductive ozonolysis.

- (A) I
- (B) II
- (C) III
- (D) IV

5. Which process of metal purification is also associated with distillation of metal

- (A) Parke's process
- (B) Amalgamation
- (C) Polling
- (D) Puddling process

6. Acetone and carbon disulphide form binary liquid solution showing positive deviation from Raoult's law with a maxima in P_T vs X graph. The normal boiling point (T_b) of pure acetone is less than that of pure CS_2 . Pick out the incorrect statements among the following.

- (A) Boiling temperature of mixture is always less than boiling temperature of acetone.
- (B) Boiling temperature of Azeotropic mixture is less than boiling temperature of pure CS_2
- (C) When a small amount CS_2 is added to excess of acetone boiling point of Resulting mixture increases.
- (D) A mixture of CS_2 and CH_3COCH_3 can be completely separated by simple fractional distillation.

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
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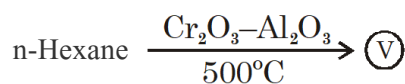
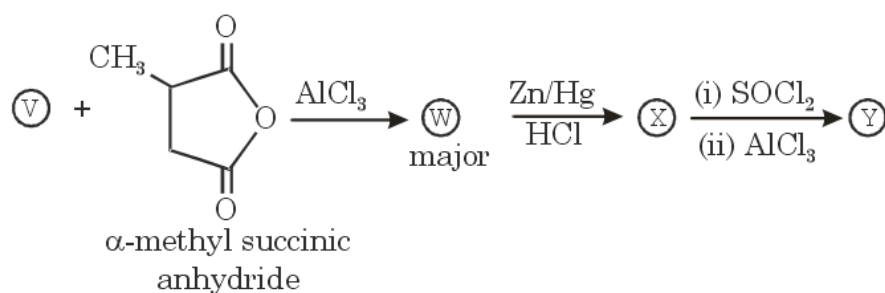
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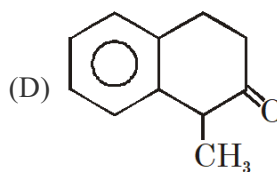
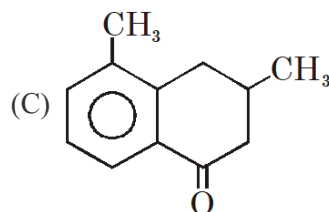
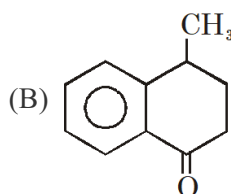
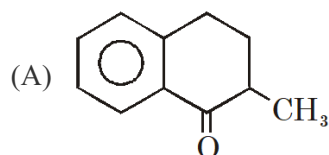
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Paragraph for Questions 7 and 8

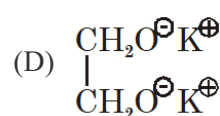
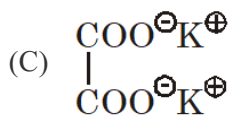
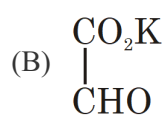
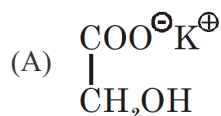
Observe the following reaction sequence and give answers to questions given below :



7. Identify (Y) :



8. If (V) $\xrightarrow[\text{H}_2\text{O}]{\text{O}_3, \text{Zn}}$ (R) $\xrightarrow{\text{Conc. KOH}}$ (S), Identify (S)



Space for Rough Work

Paragraph for Questions 9 and 10

The atomic orbitals combine to form new set of equivalent orbitals known as hybrid orbitals. Unlike pure orbitals, the hybrid orbitals are used in bond formation. The phenomenon is known as hybridisation which can be defined as the process of intermixing of the orbitals of slightly different energies in the formation of new set of orbitals of equivalent energies.

9. Which molecule is non planar as well as polar
(A) IF (B) IF_3 (C) IF_5 (D) IF_7
10. Which molecule is having same electronic geometry as well as molecular geometry
(A) NCl_3 (B) PCl_5 (C) SF_4 (D) XeF_4

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. $\text{Pt, H}_2(1 \text{ bar}) \mid 1 \text{ M NaA(aq)} \parallel 10^{-x} \text{ M HCl(aq)} \mid \text{H}_2(2 \text{ bar}), \text{Pt}$
If emf of above cell is 0.441. What is the value of x ?
(Given : $2.303 \text{ RT/F} = 0.06$; $\log 2 = 0.3$) ($\text{pK}_a(\text{HA}) = 5$)
2. How many of the following statements regarding kinetics of a chemical reaction are correct.
 - (i) The probability of collision of three or more molecules with proper orientation at same time is very rare that's why reactions having molecularity three or more are very rare.
 - (ii) In a second order reaction the number of reactant can be one but the rate will be proportional to the square of its concentration.
 - (iii) The rate of a chemical reaction at any time is always proportional to the initial concentration of the reactant molecules raised to the power equal to their order.
 - (iv) The rate of a chemical reaction also depends upon the strength of bond of reactant molecules and the number of bond to be broken.
 - (v) The total kinetic energy of the colliding molecules must be greater than some minimum value.
 - (vi) The colliding particles must be properly oriented in space when they collide.

Space for Rough Work

3. x = number of optically pure aldehyde or ketone having molecular formula of $C_5H_{10}O$ give chiral alcohol but not racemic mixture with $NaBH_4 / CH_3-OH$. (Consider enantiomer separately).
4. How many of the following compound(s) give precipitate with solution of chrome alum.
(i) $BaCl_2$ (ii) Na_2CO_3 (iii) $(NH_4)_2S$ (iv) Excess $NaOH$
5. 1 mole of an ideal gas initially present in a 2L insulated cylinder at 300 K is allowed to expand against vacuum to 8 L. then $|\Delta G_{sys}|$ in calories will be [$\ln 2 = 0.7$]
-

Space for Rough Work

6.



Number of $\text{CH}_3 - \text{I}$ consumed during the following reaction. In which Hoffmann exhaustive elimination take place.

7. For balanced chemical reaction : $'X'\text{NH}_3(\text{g}) + 'Y'\text{Cl}_2\text{O} \xrightarrow[\text{excess}]{\text{Warm}} \dots\dots\dots + 6 \text{NH}_4\text{Cl} + \dots\dots\dots$
find the value of expression $|X - Y|$.

Where, 'X' and 'Y' are number of moles of NH_3 and Cl_2O respectively required to produce six moles of NH_4Cl .

8. One mole of a real gas obeys $P(V_m - b) = RT$; where 'b' and 'R' are constants with usual meanings. If occupied volume of the gas is '10 b' then the compressibility factor of gas will be :

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

-
1. Consider the planes $P_1 : 2x + y + z + 4 = 0$, $P_2 : y - z + 4 = 0$ and $P_3 : 3x + 2y + z + 8 = 0$. Let L_1, L_2, L_3 be the lines of intersection of the planes P_2 and P_3 , P_3 and P_1 , and P_1 and P_2 respectively. Then,
 - (A) Atleast two of the lines L_1, L_2 and L_3 are non-parallel
 - (B) Atleast two of the lines L_1, L_2 and L_3 are parallel
 - (C) The three planes intersect in a line
 - (D) The three planes form a triangular prism
 2. Let $f(x)$ be a differentiable function for $x > -1$ and $f(0) = 1$, $f'(0) > 0$ also $f'(x)(1 + f(x)) = 1 + x$ then

(A) $f(x) > 0 \forall x > 0$	(B) $f(x) < 0 \forall x > 0$
(C) $f(x) > -1 \forall x > 0$	(D) $f(x) < -1 \forall x > 0$
-

Space for Rough Work

3. If the quadratic equation $x^2 + 2ax + b(a - 1) = 0$ has real roots for all real values of a , then which of the following is/are correct ?
- (A) Greatest value of b is 4 (B) No such value of b is possible
(C) Least value of b is 0 (D) b can be any integer
4. Suppose three real numbers a, b, c are in G.P. Let $z = \frac{a + ib}{c - ib}$, then
- (A) $z = \frac{ib}{c}$ (B) $z = \frac{ia}{b}$ (C) $z = \frac{ia}{c}$ (D) $z = 0$
5. In $\triangle ABC$, if $\angle B = \sec^{-1} \left(\frac{5}{4} \right) + \operatorname{cosec}^{-1} \sqrt{5}$, $\angle C = \operatorname{cosec}^{-1} \left(\frac{25}{7} \right) + \cot^{-1} \left(\frac{9}{13} \right)$ and $c = 3$.
Which of the following statement(s) is/are correct ?
(where all symbols used have their usual meanings in a triangle)
- (A) $\tan A, \tan B, \tan C$ are in arithmetic progression.
(B) Inradius the triangle is irrational
(C) Area of $\triangle ABC$ is rational.
(D) Semiperimeter is rational
6. Let $\int_0^{\pi/2^n} \ln(\sin x) dx = a_n$, $\int_0^{\pi/2^n} \ln(\cos x) dx = b_n$, then
- (A) $a_1 + b_1 = -\pi \ln 2$ (B) $a_2 + b_2 = -\frac{\pi}{2} \ln 2$
(C) $a_2 + b_2 = \frac{\pi}{2} \ln 2$ (D) $a_1 + b_1 = -\frac{\pi}{4} \ln 2$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

Paragraph for Questions 7 and 8

Let $f(x)$ be a polynomial with positive leading coefficient satisfying $f(0) = 0$ & $f(f(x)) = x \int_0^x f(t) dt \forall x \in \mathbb{R}$.

7. Two perpendicular tangents to the curve $y = f(x)$ will intersect on the curve/line
- (A) $x^2 + y^2 = 3$ (B) $y = \frac{1}{4\sqrt{3}}$ (C) $y = -\frac{\sqrt{3}}{4}$ (D) $y^2 = \frac{3x}{-16}$
8. If a line having slope 1 and passing through $(\alpha, 0)$ neither cuts nor touches the curve $y = f(x)$, then the value of α must lie in the interval.
- (A) $(0, \infty)$ (B) $\left(\frac{\sqrt{3}}{4}, \infty\right)$
- (C) $\left(\frac{3 - \sqrt{3}}{4}, \infty\right)$ (D) $\left(-\frac{3}{4}, \infty\right)$

Space for Rough Work

Paragraph for Questions 9 and 10

Let A and B are two square matrix of order 2×2 , where $\det(A) = 1$, $\det(B) = 2$, then

9. Find $\det(A + \alpha B) - \det(\alpha A + B)$, $\alpha \in \mathbb{R}$:

(A) α^2

(B) 0

(C) $\alpha^2 - 1$

(D) None of these

10. Find $\det(A - \alpha B) - \det(-\alpha A + B)$, $\alpha \in \mathbb{R}$:

(A) α^2

(B) 0

(C) $\alpha^2 - 1$

(D) None of these

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

-
1. Find the distance of the point $\hat{i} + 2\hat{j} + 3\hat{k}$ from the plane $\vec{r} \cdot (\hat{i} + \hat{j} + \hat{k}) = 5$ measured parallel to the vector $2\hat{i} + 3\hat{j} - 6\hat{k}$.
 2. Number of values of 'p' for which the equation, $(p^2 - 3p + 2)x^2 - (p^2 - 5p + 4)x + p - p^2 = 0$ possess more than two roots, is ?
 3. Area of triangle formed by the asymptotes and any tangent to $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is 36 sq. units, where a & b are integers, then number of such hyperbolas is ?
-

Space for Rough Work

4. From an external point P, two tangents are drawn to the parabola $x^2 = -8y$. If chord of contact of these tangents is tangential to the circle $x^2 + y^2 = 4$, then locus of P will be a hyperbola length of whose transeverse axis is ?
5. All the face cards from a pack of 52 playing cards are removed. From the remaining pack half of the cards are randomly removed whithout looking at them and then randomly drawn two cards simultaneously from the remaining. If the probability that, two cards drawn are both aces, is $\frac{p \binom{38}{20}}{40C_{20} \cdot 20C_2}$, then the value of p is ?
-

Space for Rough Work

6. Normal to the curve $y = f(x)$ at $(1, 1)$ is $3x + 4y = 7$. Also $f(x) \geq \frac{4x-1}{3}$ for $x \geq 1$ and $f(x) \leq \frac{4x-1}{3}$ for $x \leq 1$. where $f(x)$ is twice differentiable everywhere in its domain. Evaluate $\lim_{x \rightarrow 1} \frac{3f'(x) - 2f(x) - 2x}{f(x) - x^2}$.
7. Using only 0, 1, 2, 3, 4, number of all four digit numbers whose sum of digit is 12, is
8. If $n > 3$, then $\binom{n-1}{1}(a-1)^2 - \binom{n-1}{2}(a-2)^2 + \binom{n-1}{3}(a-3)^2 - \dots + (-1)^{n-2} \binom{n-1}{n-1}(a-n+1)^2 = k$
then number of positive divisors of k , (given 'a' is a prime number)

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

Space for Rough Work